microct-porous-media Scientific User Manual

Version 1.0.0

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A Python package for X-ray microcomputed tomography (µCT) image analysis of soils and horticultural growing media.

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Table of Contents

1. Introduction

microct-porous-media is a Python package for processing and analysing three-dimensional X-ray microcomputed tomography (µCT) images of soils and horticultural substrates. The package was migrated from six original research scripts and organizes them into importable modules that can be reused in scientific workflows.

The package is designed for researchers who want to extract pore-structure and hydraulic information directly from reconstructed µCT image stacks. It supports workflows from image filtering and segmentation to REV analysis, pore morphology, pore-network extraction, water-retention estimation, permeability and saturated hydraulic conductivity calculation.

1.1 Scientific Applications

- Representative Elementary Volume (REV) analysis for visible porosity.
- Quantification of pore morphology and topology.
- Pore size distribution and image-based water retention curve estimation.
- Critical pore diameter and percolating throat analysis.
- Pore network modelling with PoreSpy and OpenPNM.
- Saturated hydraulic conductivity estimation using Stokes flow and Darcy-based conversion.

1.2 Modules

Module	Purpose
rev_xy.py	2D REV analysis by subdividing the X-Y plane while retaining full image depth.
rev_xyz.py	3D REV analysis by subdividing X, Y and Z directions simultaneously.
pore_metrics.py	Pore morphology, topology, percolation and network metrics.
critical_pore.py	Critical throat or pore diameter from a percolating network.
pore_size_distribution.py	Local-thickness pore size distribution and simplified water retention curve.
permeability.py	Pore network permeability and saturated hydraulic conductivity.

2. Scientific Background

2.1 µCT Image Analysis

µCT provides non-destructive three-dimensional images of the pore space of soils and growing media. Reconstructed grayscale image stacks are segmented into pore and solid phases. Once binary segmentation is available, the voxel structure can be used to calculate porosity, pore morphology, topology and network-based hydraulic properties.

2.2 Representative Elementary Volume

The Representative Elementary Volume (REV) is the minimum volume at which a measured property becomes statistically stable. In this package, REV is evaluated using visible pore volume fraction. Stability is assessed using mean porosity, standard deviation and coefficient of variation across subvolumes.

2.3 Visible Porosity

Visible porosity is calculated from binary images as the number of pore voxels divided by the total number of voxels. It is resolution dependent: pores smaller than the voxel size cannot be resolved and therefore do not contribute directly to visible porosity.

2.4 Pore Network Modelling

Pore network modelling approximates the pore space as pores connected by throats. The package uses PoreSpy SNOW2 extraction and OpenPNM network representation. This allows pore and throat metrics as well as pressure-driven Stokes flow simulations.

2.5 Hydraulic Conductivity and Permeability

Permeability is an intrinsic property of a porous medium, while saturated hydraulic conductivity also depends on fluid density, viscosity and gravity. The package calculates permeability from flow rate, sample length, cross-sectional area, fluid viscosity and pressure drop. Hydraulic conductivity is then derived from permeability.

```
K_abs = Q * L * mu / (A * Delta_P)
Ksat = K_abs * (rho * g / mu)
```

3. Installation

3.1 Recommended Python Version

Use Python 3.10, 3.11 or 3.12. Python 3.14 is not recommended because several scientific dependencies such as scikit-image, PoreSpy, OpenPNM and numba may not yet provide stable binary packages for that version.

3.2 Python Virtual Environment

```
cd D:\Python Package_Image\microctpm_full_package
py -3.11 -m venv venv311
venv311\Scripts\activate
python -m pip install --upgrade pip
pip install -e .
python -c "import microctpm; print(microctpm.__version__)"
```

3.3 Conda Environment

```
conda env create -f environment.yml
conda activate microctpm
pip install -e .
```

4. Input Data Requirements

Item	Requirement
Image format	.tif or .tiff 3D grayscale image stack.
Dimensions	Any 3D image shape is supported if memory is sufficient, e.g. (435,435,435), (1000,1000,1000), or non-cubic stacks such as (601,1100,1100).
Voxel size	Must be supplied or adjusted according to the scan resolution. Typical examples are 15 µm and 46 µm.
Segmentation threshold	Must be selected for each dataset. Manual, Otsu global thresholding and local thresholding are possible.
Memory	Large volumes can require substantial RAM, especially for PoreSpy/OpenPNM workflows.

5. Threshold Selection and Segmentation

Threshold selection is one of the most important steps in the entire package. All subsequent calculations depend on whether pore and solid voxels are correctly separated.

5.1 Manual Thresholding

Manual thresholding uses a user-defined grayscale value. It is appropriate when the image histogram and material contrast are known. The example thresholds in the code, such as 55, 85 or 110, are dataset-specific and not universal.

```
binary_image = image < fixed_threshold
```

5.2 Global Otsu Thresholding

For automatic global segmentation, users may compute a threshold using Otsu method from scikit-image and pass the resulting value to the package functions.

```
from skimage.filters import threshold_otsu
import tifffile

image = tifffile.imread("sample.tif")
threshold = threshold_otsu(image)
print(threshold)
```

5.3 Local or Adaptive Thresholding

Local thresholding may be useful when illumination, contrast or material composition varies spatially. It is more computationally expensive and parameter dependent.

```
from skimage.filters import threshold_local

local_threshold = threshold_local(image_slice, block_size=51)
binary_slice = image_slice < local_threshold
```

5.4 Recommended Workflow

1. Inspect image slices and the grayscale histogram.
2. Compute an Otsu threshold as an objective reference.
3. Compare manual and Otsu-based segmentation visually.
4. Validate visible porosity against independent measurements when available.
5. Document the selected threshold and method in publications.

6. Package Structure

```
microctpm_full_package/
├── src/microctpm/
│   ├── rev_xy.py
│   ├── rev_xyz.py
│   ├── pore_metrics.py
│   ├── critical_pore.py
│   ├── pore_size_distribution.py
│   └── permeability.py
├── examples/
├── tests/
├── dist/
├── README.md
├── pyproject.toml
├── requirements.txt
├── environment.yml
└── LICENSE
```

7. Module: REV Analysis in XY Plane (rev_xy.py)

This module performs 2D REV analysis by dividing the X-Y plane while retaining the full Z depth. This strategy is useful for horticultural substrates where vertical gradients or packing effects may make full 3D subdivision less robust.

7.1 Workflow

6. Load 3D µCT image.
7. Apply Gaussian filter with sigma = 1.0.
8. Segment pore space using a fixed/manual threshold.
9. Divide the X-Y plane into n x n tiles.
10. Calculate visible porosity for each subvolume.
11. Calculate mean, standard deviation and coefficient of variation.
12. Export CSV files and REV plot.

7.2 Functions

Function	Description
load_and_filter_image(image_path)	Loads a 3D TIFF image and applies Gaussian filtering.
create_subvolumes(image, fixed_threshold=55)	Segments the image and creates X-Y subvolumes.
compute_pore_volume_metrics(subvolumes, positions, voxel_size, output_dir)	Calculates porosity statistics and saves CSV/figure outputs.
run_rev_xy(image_path, output_dir, voxel_size=46, fixed_threshold=55)	Runs the full REV XY workflow.

7.3 Parameters

Parameter	Meaning	Typical value
image_path	Path to 3D TIFF image.	sample.tif
output_dir	Folder for CSV and figure outputs.	results
voxel_size	Voxel size in µm.	15 or 46
fixed_threshold	Manual segmentation threshold.	Dataset-specific

7.4 Division Steps

The module uses subdivision levels: 2, 3, 4, 5, 6, 7, 8, 9, 10, 15, 20, 25, 30, 35, 40, 45 and 50. For each n, n x n subvolumes are generated. The full Z depth is retained.

7.5 Example

```
from microctpm.rev_xy import run_rev_xy

results = run_rev_xy(
    image_path="sample.tif",
    output_dir="results",
    voxel_size=15,
    fixed_threshold=85
)
```

7.6 Outputs

Output	Content
RHP_1_fullxy_raw_1.csv	Subvolume porosity and physical size for each tile.
coefficient_of_variation_RHP_1_cv.csv	Mean porosity, standard deviation and CV for each subdivision level.
RHP_1_xydivition.tiff	REV plot with porosity and coefficient of variation.

8. Module: REV Analysis in XYZ Space (rev_xyz.py)

This module performs full 3D REV analysis by subdividing the Z, Y and X axes simultaneously. For each subdivision level n, it creates n x n x n subvolumes. This analysis is more computationally expensive than REV XY and is often better suited to mineral soils or relatively homogeneous porous media.

8.1 Workflow

13. Load and filter the 3D image.
14. Segment pore space using a threshold.
15. Divide Z, Y and X into equal segments.
16. Calculate visible porosity for each cube-like subvolume.
17. Calculate mean, standard deviation and CV.
18. Save CSV and TIFF outputs.

8.2 Functions

Function	Description
load_and_filter_image(image_path)	Loads and Gaussian-filters the image.
create_subvolumes(image, fixed_threshold=85)	Creates 3D subvolumes after segmentation.
compute_pore_volume_metrics(subvolumes, positions, voxel_size, output_dir)	Computes REV metrics.
run_rev_xyz(image_path, output_dir, voxel_size=15, fixed_threshold=85)	Runs full 3D REV workflow.

8.3 Example

```
from microctpm.rev_xyz import run_rev_xyz

results = run_rev_xyz(
    image_path="sample.tif",
    output_dir="results",
    voxel_size=15,
    fixed_threshold=85
)
```

8.4 Outputs

Output	Content
subvolume_pore_volume_fullxyz_WF4raw_cv_1.csv	Raw subvolume porosity values and positions.
coefficient_of_variation_WF4raw_1.csv	CV statistics for each subdivision level.
pore_volume_and_cv_vs_size_full_PeatS_1_cv_3D.tiff	3D REV plot.

8.5 Interpretation

REV is indicated by stabilization of mean visible porosity and reduction of CV. REV XYZ is stricter than REV XY because heterogeneity in all three directions contributes to variability.

9. Module: Pore Metrics Analysis (pore_metrics.py)

pore_metrics.py is the central morphology and topology module. It calculates pore volume, pore area, pore size measures, sphericity, surface density, mean curvature density, percolation, pore class volumes, Euler number density, gamma indicator and network properties using PoreSpy and OpenPNM.

9.1 Main Workflow

19. Load image and apply Gaussian filtering.
20. Segment using fixed/manual threshold.
21. Label connected pore regions.

22. Calculate pore volume, pore area, equivalent diameters and sphericity.
23. Classify pore voxels into macro-, meso- and micropores.
24. Compute Euler number density and gamma indicator.
25. Clean binary image using remove_small_objects and remove_small_holes.
26. Extract pore network with PoreSpy SNOW2.
27. Create OpenPNM network and calculate pore/throat metrics.
28. Export CSV, figures and Excel summary.

9.2 Functions

Function	Description
load_and_filter_image(image_path)	Loads 3D TIFF and applies Gaussian smoothing with sigma=2.
compute_euler_number(binary_image)	Calculates Euler number density from connected components and skeleton voxels.
compute_gamma_indicator(binary_image)	Calculates gamma indicator from connected pore cluster sizes.
analyze_pore_network(binary_image)	Extracts PoreSpy SNOW2 network and OpenPNM metrics.
analyze_pore_properties(binary_image)	Calculates pore volume, surface, diameters, sphericity and pore classes.
analyze_pore_structure(binary_image, original_image)	Combines morphology, topology, threshold histogram and network analysis.
compute_metrics(image)	Main user function that returns a pandas DataFrame and writes output files.

9.3 Key Metrics

Metric	Meaning
Pore Volume (μm³)	Total volume of segmented pore voxels.
Total Pore Area (mm²)	Estimated surface area of pore boundaries.
Max/Min/Avg Pore Diameter (μm)	Equivalent pore diameter from labelled pore regions or distance maps.
Avg Pore Sphericity	Shape descriptor indicating closeness to a sphere.
Pore Surface Density (mm⁻¹)	Pore surface area normalized by sample volume.
Mean Curvature Density (mm⁻²)	Curvature-based descriptor of pore-solid interface complexity.
Percolation	Whether pore space connects across the image domain.
Critical Pore Diameter (μm)	Limiting pore diameter computed in the morphology workflow.
Chi Density (mm⁻³)	Euler number density; more negative values indicate higher connectivity.
Gamma	Indicator of pore cluster dominance/connectivity, close to 1 for a dominant connected pore cluster.
Num Pores / Num Throats	Pore network size from PoreSpy/OpenPNM.
Avg Coordination Number	Average number of neighboring pores connected by throats.
Avg Throat Diameter (μm)	Mean pore throat diameter from network extraction.
Phi_vis (%)	Visible porosity percentage from the segmented image.

9.4 Pore Classes

The module classifies pore voxels according to pore size: macropores > 300 μm, mesopores 50-300 μm and micropores < 50 μm. These thresholds are practical categories and should be interpreted relative to image resolution and research objective.

9.5 Example

```
from microctpm.pore_metrics import load_and_filter_image, compute_metrics

image = load_and_filter_image("sample.tif")
metrics = compute_metrics(image)
print(metrics)
```

9.6 Output Files

File	Description
pore_properties_raw.csv	Raw pore region properties such as diameter and shape descriptors.
pore_sizes_raw.csv	Voxel-based pore size data.
pore_diameters.csv	Large pore diameter table written in chunks for memory efficiency.
pore_size_distribution.csv	Binned pore size distribution.
throat_radii.csv	Extracted network throat radii.
analysis_results.xlsx	Final metrics table.
pore_size_distribution.png	Pore size distribution plot.
thresholded_histogram.png	Pore and solid intensity histograms with threshold.
throat_size_distribution.png	Throat size distribution plot.

9.7 Interpretation

High visible porosity indicates a large fraction of resolved pore space. A strongly negative Euler number density indicates many connected loops and a highly connected pore network. Gamma values close to 1 indicate that most pore voxels belong to a dominant connected cluster. High coordination number indicates many pore-to-pore connections and can be associated with increased drainage and hydraulic conductivity.

9.8 Warnings

FutureWarning messages from scikit-image may appear because some functions have changed names or parameters in recent versions. These warnings do not necessarily indicate failure. PoreSpy/OpenPNM warnings about unused physical properties may also appear and are not fatal unless the target metric depends on the missing property.

10. Module: Critical Pore Diameter (critical_pore.py)

This module estimates the critical pore or throat diameter along a percolating path through the pore network. It uses PoreSpy SNOW2 to extract a network and graph methods to evaluate inlet-to-outlet connectivity.

10.1 Functions

Function	Description
trim_disconnected_pores(network)	Retains the largest connected pore component.
calculate_critical_pore_diameter(image_path, voxel_size=15e-6, threshold=85, r_max=10, sigma=1)	Runs the complete critical pore diameter workflow.

10.2 Parameters

Parameter	Meaning
image_path	Path to 3D TIFF image.
voxel_size	Voxel size in metres, e.g. 15e-6.
threshold	Segmentation threshold.
r_max	Maximum radius parameter used in local-thickness/network processing.
sigma	Gaussian filter sigma.

10.3 Example

```
from microctpm.critical_pore import calculate_critical_pore_diameter

cpd = calculate_critical_pore_diameter(
    image_path="sample.tif",
    voxel_size=15e-6,
    threshold=85,
```

```
        r_max=10,
        sigma=1
    )
    print(cpd)
```

10.4 Interpretation

The critical pore diameter represents a limiting size along a connected flow pathway. Larger critical diameters generally indicate more open percolating pathways and potentially higher drainage or conductivity. However, it should be interpreted together with porosity, connectivity and throat size distribution.

11. Module: Pore Size Distribution and WRC (pore_size_distribution.py)

This module estimates pore volumes, pore diameters and a simplified image-based water retention curve. It uses the local-thickness concept to infer pore size from the segmented pore space.

11.1 Functions

Function	Description
load_and_filter_image(image_path)	Loads and filters a 3D TIFF image.
compute_pore_volumes_and_sizes(image)	Segments pores, applies local thickness, computes pore volumes/sizes and WRC data.

11.2 Outputs

Output	Content
pore_size_distribution_wrc.xlsx	Excel file containing metrics, pore data and WRC data.
metrics dictionary	Total pore volume, total sample volume, number of pores, average pore volume and average pore size.
pore_data DataFrame	Single-pore volume and equivalent pore size.
wrc_data DataFrame	Pore size, matric potential and volumetric water content.

11.3 Example

```
from microctpm.pore_size_distribution import load_and_filter_image, compute_pore_volumes_and_sizes

image = load_and_filter_image("sample.tif")
metrics, pore_data, wrc_data = compute_pore_volumes_and_sizes(image)
print(metrics)
print(wrc_data.head())
```

11.4 Interpretation

Large pores drain at lower matric suction, while smaller pores retain water at higher suction. The image-based WRC is resolution-limited because pores smaller than the voxel size cannot be resolved. Users should compare the resulting WRC with laboratory measurements whenever possible.

12. Module: Permeability and Saturated Hydraulic Conductivity (permeability.py)

This module estimates intrinsic permeability and saturated hydraulic conductivity by extracting a pore network, assigning hydraulic conductance and solving Stokes flow using OpenPNM.

12.1 Workflow

29. Load and Gaussian-filter image.
30. Segment pores by threshold.
31. Optionally save image histogram and binary slice.

32. Pad image for SNOW2 extraction.
33. Extract network with PoreSpy.
34. Convert to OpenPNM network.
35. Trim disconnected components.
36. Scale geometry by voxel size.
37. Add geometry and hydraulic conductance models.
38. Define inlet and outlet pores.
39. Solve Stokes flow.
40. Compute permeability and hydraulic conductivity.
41. Optionally save 3D network/pressure figure.

12.2 Function

Function	Description
calculate_permeability(...)	Main workflow for permeability and Ksat calculation.
calculate_hydraulic_conductivity(...)	Alias for calculate_permeability.
trim_disconnected_pores(network)	Keeps the largest connected network component.

12.3 Important Parameters

Parameter	Description	Example
image_path	Path to 3D TIFF image.	sample.tif
output_dir	Output folder.	results
voxel_size	Voxel size in metres.	15e-6
threshold	Segmentation threshold.	85
sigma	Gaussian filter sigma.	1
boundary_width	SNOW2 boundary width.	5
inlet_pressure	Inlet pressure in Pa.	2000
outlet_pressure	Outlet pressure in Pa.	1990
pressure_drop	Pressure drop used in Darcy-law permeability calculation.	1990
viscosity	Dynamic viscosity of water in Pa s.	1.0e-3
density	Water density in kg m-3.	1000
gravity	Gravitational acceleration in m s-2.	9.81
flow_axis	Flow axis: 0, 1 or 2.	0

12.4 Pressure Drop

The pressure boundary conditions used by the flow solver and the pressure drop used in Darcy-law conversion should be handled carefully. In your validated workflow, pressure_drop=1990 Pa reproduced the original script convention and produced realistic hydraulic conductivity for the tested image. If pressure_drop is omitted, the package may use abs(inlet_pressure - outlet_pressure), depending on the implementation. Users must document the selected pressure-drop convention.

Pressure Boundary Conditions and Pressure Drop

The permeability calculation is based on a pressure gradient imposed across the extracted pore network.

Users may specify:

inlet_pressure

outlet_pressure

pressure_drop

The pressure boundary conditions define the hydraulic gradient that drives flow through the pore network.

If:

pressure_drop = None

the package automatically calculates:

$\Delta P = |P\_inlet - P\_outlet|$

Example:

inlet_pressure = 2000 Pa

outlet_pressure = 1990 Pa

gives:

$\Delta P = 10$ Pa

Important Scientific Note

Pressure drop is not a universal constant.

The appropriate pressure difference depends on:

pore size distribution

throat size distribution

network connectivity

hydraulic gradient

experimental conditions

imposed boundary conditions

Therefore, a pressure difference used for one porous medium should not automatically be applied to another porous medium.

The values shown in examples are intended only to demonstrate package usage and should not be interpreted as universal settings.

Recommended Practice

For general use:

pressure_drop = None

For research applications:

Use pressure boundary conditions that correspond to the experimental or modelling scenario being investigated.

Always report:

inlet_pressure

outlet_pressure

pressure_drop

when publishing permeability or hydraulic conductivity results.

12.5 Example

12.5 Example

from microctpm.permeability import calculate_permeability

results = calculate_permeability(
image_path="sample.tif",
output_dir="results",
voxel_size=15e-6,
threshold=85,
inlet_pressure=2000.0,
outlet_pressure=1990.0,
pressure_drop=None,
save_outputs=True,
make_plot=True,
show_plot=False
)

print(results)

12.6 Outputs

Output	Meaning
porosity	Visible porosity of segmented image.
flow_rate_m3_s	Simulated flow rate.
permeability_m2	Intrinsic permeability in square metres.
permeability_mD	Permeability in millidarcy.
hydraulic_conductivity_cm_s	Saturated hydraulic conductivity in cm/s.
num_pores / num_throats	OpenPNM network size.
num_inlet_pores / num_outlet_pores	Boundary pores used by Stokes flow.
pressure_drop_Pa	Pressure drop used in Darcy-law calculation.
intensity_histogram.tif	Optional grayscale histogram.
binary_image_slice.tif	Optional segmented slice.
3d_pore_network.tif	Optional network/pressure visualization.

13. Complete Example Workflow

The following workflow demonstrates how a user can analyse a 3D µCT image named sample.tif. Replace voxel size and threshold with values appropriate for the dataset.

```
# 1. REV XY
from microctpm.rev_xy import run_rev_xy
run_rev_xy("sample.tif", "results/rev_xy", voxel_size=15, fixed_threshold=85)

# 2. REV XYZ
from microctpm.rev_xyz import run_rev_xyz
run_rev_xyz("sample.tif", "results/rev_xyz", voxel_size=15, fixed_threshold=85)

# 3. Pore metrics
from microctpm.pore_metrics import load_and_filter_image, compute_metrics
image = load_and_filter_image("sample.tif")
metrics = compute_metrics(image)
```

```
# 4. Critical pore diameter
from microctpm.critical_pore import calculate_critical_pore_diameter
cpd = calculate_critical_pore_diameter("sample.tif", voxel_size=15e-6, threshold=85)

# 5. Pore size distribution and WRC
from microctpm.pore_size_distribution import load_and_filter_image, compute_pore_volumes_and_sizes
image = load_and_filter_image("sample.tif")
metrics, pore_data, wrc_data = compute_pore_volumes_and_sizes(image)

# 6. Permeability and Ksat
from microctpm.permeability import calculate_permeability
k = calculate_permeability("sample.tif", output_dir="results/permeability", voxel_size=15e-6,
threshold=85, pressure_drop=1990.0)
```

14. Validation Example

During package testing, a real cropped 3D µCT image was processed through all six modules. The exact values are dataset-specific and should be treated as validation evidence for the tested image, not universal expected results.

Metric	Validated example value
Image size	435 x 435 x 435 voxels
Voxel size	15 µm
Threshold	85
Visible porosity from pore metrics	64.30 %
Critical pore diameter	222.62 µm
Pore network size	15,649 pores and 55,685 throats in pore_metrics workflow
Permeability	3.16e-11 m²
Permeability	32,002 mD
Hydraulic conductivity	0.0310 cm/s
Pressure drop used	1990 Pa

15. Troubleshooting

Problem	Likely cause	Solution
ModuleNotFoundError: microctpm	Package not installed or wrong environment.	Activate the correct environment and run pip install -e .
Python 3.14 installation failure	Scientific dependencies unavailable for very new Python version.	Use Python 3.10-3.12, preferably Python 3.11.
FileNotFoundError for output folder	Output directory does not exist.	Create output folder or use a function that creates it automatically.
local_thickness unexpected keyword mode	PoreSpy API version mismatch.	Remove unsupported mode argument or use compatible PoreSpy version.
Tkinter or FigureCanvasAgg warning	Non-interactive plotting backend.	Use show_plot=False and save figures to files.
OpenPNM warnings about surface_tension or diffusivity	Models not needed for hydraulic conductance were skipped.	Usually safe to ignore unless those properties are required.
Very slow SNOW2 extraction	Image too large or high porosity network.	Crop image, increase RAM, or process REV-sized subvolume.
Unrealistically high Ksat	Pressure drop convention or segmentation threshold may be wrong.	Check pressure_drop, inlet/outlet pressures, threshold and connectivity.

16. References and Citation

Primary scientific reference for the methodology:

Muhammed, H. H.; Anlauf, R.; Daum, D.; Schmitt Rahner, M.; Kuka, K. (2025). Pore Structure Analysis of Growing Media Using X-Ray Microcomputed Tomography. Applied Sciences, 15(22), 11886. DOI: 10.3390/app152211886

16.1 Suggested Package Citation

```
@software{microct_porous_media_2026,
  author = {Muhammed, Hadi Hamaaziz and Anlauf, Ruediger},
  title = {microct-porous-media: Python tools for µCT image analysis of soils and horticultural substrates},
  version = {1.0.0},
  year = {2026}
}
```

16.2 Paper BibTeX

```
@article{Muhammed2025MicroCT,
  author = {Muhammed, Hadi Hamaaziz and Anlauf, Ruediger and Daum, Diemo and Schmitt Rahner, Mayka and Kuka, Katrin},
  title = {Pore Structure Analysis of Growing Media Using X-Ray Microcomputed Tomography},
  journal = {Applied Sciences},
  year = {2025},
  volume = {15},
  number = {22},
  pages = {11886},
  doi = {10.3390/app152211886}
}
```

Appendix A: Function Reference

Module	Public functions
rev_xy.py	load_and_filter_image; create_subvolumes; compute_pore_volume_metrics; run_rev_xy
rev_xyz.py	load_and_filter_image; create_subvolumes; compute_pore_volume_metrics; run_rev_xyz
pore_metrics.py	load_and_filter_image; compute_euler_number; compute_gamma_indicator; analyze_pore_network; analyze_pore_properties; analyze_pore_structure; compute_metrics
critical_pore.py	trim_disconnected_pores; calculate_critical_pore_diameter
pore_size_distribution.py	load_and_filter_image; compute_pore_volumes_and_sizes
permeability.py	trim_disconnected_pores; calculate_permeability; calculate_hydraulic_conductivity

Appendix B: Output File Checklist

- Always check CSV and Excel files after each run.
- Compare visible porosity with expected or laboratory porosity when available.
- Inspect saved plots for axis labels, threshold quality and unexpected distributions.
- Record threshold, voxel size, image dimensions and preprocessing settings for reproducibility.
- For permeability, always report pressure boundary conditions and pressure_drop used in Darcy conversion.